



Enhancing Regional Machine Learning Weather Prediction Using Tailored Loss Functions and High-Resolution RRJ-ClimCORE Reanalysis

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Abstract

Recent advances in machine learning-based weather prediction (MLWP) have achieved accuracy comparable to operational numerical weather prediction systems, while offering much faster and more energy-efficient inference. However, global MLWP models struggle to forecast localized extremes, including typhoons and heavy rainfall. This is mainly due to coarse resolution of training data and the reliance on mean squared error (MSE) loss, which inclines toward spatial smoothing. This study develops a regional MLWP model for Japan trained on high-resolution regional reanalysis data with the employment of variable-specific loss functions for wind, pressure, geopotential height, and precipitation. Ablation experiments demonstrate that these loss functions contribute in complementary ways, substantially improving the forecast skill relative to conventional MSE loss. Case studies of an extratropical cyclone and Typhoon Nanmadol show improved forecasts of cyclone intensity, strong winds, and orographic rainfall. Leveraging regional reanalysis with the tailored loss design is therefore an effective strategy for regional MLWP.

Keywords Machine learning weather prediction · Regional forecast model · Regional reanalysis · Loss function · Extreme event forecasting

1 Introduction

In recent years, machine learning-based weather prediction (MLWP) has gained increasing attention. Many global MLWP models (Pathak et al. 2022; Lam et al. 2023; Bi et al. 2023; Chen et al. 2023, 2025; Lang et al. 2024) now approach the skill of operational numerical weather

prediction (NWP) systems (Rasp et al. 2024) while running orders of magnitude faster with far less energy use. These global MLWP models are generally trained to learn the temporal evolution patterns of multiple meteorological variables. Because temporal consistency in the quality of the training data is essential, global reanalysis datasets, such as ECMWF Reanalysis v5 (ERA5) (Hersbach et al. 2020), are commonly used.

Despite this progress, MLWP models still face significant challenges in forecasting localized and extreme weather events. For example, recent evaluations show that global MLWP models tend to yield smaller track errors but systematically underestimate tropical cyclone intensity, including positive biases in minimum sea-level pressure (SLP) (DeMaria et al. 2025; Sahu et al. 2025; Yamaguchi et al. 2025). Similarly, forecasts of predicted strong winds or heavy precipitations are often oversmoothed, failing to capture their locally intense structures. Such limitations are particularly problematic for disaster-mitigation purposes and regional applications.

These shortcomings appear across diverse MLWP neural-network architectures, stemming mainly from (1) the training data and (2) training methodology. With respect to (1), most

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global MLWP models rely on global reanalyses with horizontal grid-spacing exceeding 20 km, which misses detailed topographic features, leaving orographic precipitation and organized mesoscale convection unresolved. Regarding (2), the prevalent use of mean squared error (MSE) as the loss function has several drawbacks, such as the double-penalty effect: when the forecast misplaces a localized event, the model is penalized twice—once at the forecasted location (a false alarm where the ground truth is small) and additionally at the ground-truth location (a miss where the forecast lacks the event), thereby inflating the loss and thus causing the model to smooth out extremes (Hoffman et al. 1995; Ebert et al. 2013). More generally, using the same MSE loss form for all variables—while per-variable weights, as is standard practice, can adjust their relative importance—does not explicitly account for variable-specific physical characteristics, such as differences in spatial smoothness, localization, or whether a field is scalar or vector.

To address (1) the training-data limitation, we train a regional MLWP model over Japan and surrounding areas (Fig. S1) using RRJ-ClimCORE (Nakamura et al. 2022), a state-of-the-art high-resolution regional reanalysis. RRJ-ClimCORE is being produced by assimilating diverse observations—largely the same as those used in operations at the Japan Meteorological Agency (JMA), with selected components reprocessed—into JMA’s mesoscale analysis system. With a horizontal grid spacing of 5 km, it successfully resolves regional phenomena that global reanalyses cannot. This is particularly valuable for Japan, where steep and complex terrain strongly modulates atmospheric fields, and where typhoons and organized mesoscale convective systems influence frequently.

To address (2) the methodological limitation, we modify the design of loss functions. While recent studies have introduced regional MLWP models (Oskarsson et al. 2023; Larsson et al. 2025; Xu et al. 2025; Qin et al. 2024; Adamov et al. 2025) or hybrid approaches that can flexibly switch between global and regional domains (Nipen et al. 2024; Wijnands et al. 2025), these models adopt a single loss formulation across meteorological variables—in practice typically a weighted MSE. To better reflect the distinct properties of individual prognostic variables, our MLWP model employs variable-specific loss functions for wind, pressure, geopotential height, and precipitation, thereby incorporating their physical properties more explicitly into the learning process. This aims to improve variable representation and, consequently, improve extreme-event forecasts. To assess model performance from multiple perspectives, we conducted fuzzy verification for precipitation—a coarser-scale evaluation that tolerates small space-time offsets—as well as conventional evaluations based on root-mean-square error (RMSE). We also conducted case studies of localized

extreme events, including heavy precipitation and strong winds. Our experiments demonstrate that refining the loss function can substantially improve forecasts of extreme events.

These findings highlight that for achieving reliable quantitative predictions of regional extreme events, not only model sophistication but also carefully tailored loss function design is essential.

2 Methodology

2.1 Dataset: Trial Version of RRJ-ClimCORE

RRJ-ClimCORE (Nakamura et al. 2022) is a regional reanalysis dataset for Japan and its surrounding regions, developed jointly by the University of Tokyo and JMA. It is produced using the March 2022 operational version of JMA mesoscale forecast system based on a nonhydrostatic regional model with 5-km horizontal grid spacing and 96 vertical levels with a mesoscale 4D-Var data assimilation system. By assimilating diverse observations (e.g., satellite, radar/rain-gauge, conventional), RRJ-ClimCORE provides atmospheric fields of operational quality. While the final product will use lateral boundary conditions derived from the JRA-3Q long-term global reanalysis (Kosaka et al. 2024), in this study, we use a 4.5-year trial dataset (July 2018–December 2022) based on boundary conditions from JMA’s operational global forecasts.

From the RRJ-ClimCORE data, we extracted an $N \times M = 470 \times 470$ grid domain (Fig. S1) at 5 km grid spacing and selected a set of prognostic variables $\mathcal{V}$ (Table 1; $|\mathcal{V}| = 28$). At time t , these variables on the $N \times M$ grid form the 3-D state tensor $\mathbf{X}^t = (X_{\alpha nm}^t)_{\alpha \in \mathcal{V}, 0 \leq n < N, 0 \leq m < M} \in \mathbb{R}^{|\mathcal{V}| \times N \times M}$, with $\alpha \in \mathcal{V}$ indexing variables, n, m indexing grid points in the x - and y -directions. In addition, we assembled ancillary predictor fields F^t that are not included in $\mathcal{V}$: static geographic fields (surface elevation, land–sea mask, latitude, longitude) and prescribed time-varying fields (top-of-the-atmosphere downward shortwave radiation and sea-surface temperature). F^t are used as prescribed model inputs and are not prognostic outputs.

2.2 Model: Hi-LAM

As the regional MLWP model, we used Hi-LAM with default hyperparameters, available in open-source repository Neural-LAM (Oskarsson et al. 2023). Hi-LAM is a graph neural network based model with a hierarchical mesh structure.

Table 1 List of prognostic variables

Type	Variable	Short name α	Weight w_α
Surface	1.5 m Temperature	SAT	1.0
	10 m U-component of wind	Usfc	0.1
	10 m V-component of wind	Vsfc	0.1
	Sea level pressure	SLP	0.1
	1-hourly precipitation	R1h	0.1
	Total precipitable water	TPW	0.1
	Downward longwave radiation	DLR	0.1
	Downward shortwave radiation	DSR	0.1
	Temperature	T ₁₀₀₀ , T ₈₅₀ , T ₅₀₀ , T ₃₀₀	1000 hPa: 0.1
Pressure level (1000, 850, 500, 300 hPa)	U-component of wind	U ₁₀₀₀ , U ₈₅₀ , U ₅₀₀ , U ₃₀₀	850 hPa: 0.05
	V-component of wind	V ₁₀₀₀ , V ₈₅₀ , V ₅₀₀ , V ₃₀₀	500 hPa: 0.03
	Specific humidity	Q ₁₀₀₀ , Q ₈₅₀ , Q ₅₀₀ , Q ₃₀₀	300 hPa: 0.02
	Geopotential height	Z ₁₀₀₀ , Z ₈₅₀ , Z ₅₀₀ , Z ₃₀₀	

In total, $|\mathcal{V}| = 28$ variables are included, and the short names are also used as indices $\alpha \in \mathcal{V}$

Forecasts are produced as discrete, hourly, second-order autoregressive updates. At each step, Hi-LAM ingests $(\mathbf{X}^{t-2}, \mathbf{X}^{t-1}, F^t)$ and outputs the next state, denoted $\hat{\mathbf{X}}^t$; multi-step forecasts are obtained by iterating this update (rollout).

2.3 Design of Loss Functions

We aim to improve forecast performance by designing loss functions that can account for the physical properties of individual prognostic variables. Specifically, we combine four types of loss functions:

$$\mathcal{L}_{\text{total}} = \frac{1}{N_{\text{rollout}}} \sum_{t \in \mathbb{T}} \left[\lambda_{\text{mse}} \mathcal{L}_{\text{mse}}^{\mathcal{V}_{\text{mse}}}(\hat{\mathbf{X}}^t, \mathbf{X}^t) + \lambda_{\text{wind}} \mathcal{L}_{\text{wind}}(\hat{\mathbf{X}}^t, \mathbf{X}^t) \right. \\ \left. + \lambda_{\text{falc}} \mathcal{L}_{\text{falc}}^{\mathcal{V}_{\text{falc}}}(\hat{\mathbf{X}}^t, \mathbf{X}^t) + \lambda_{\text{grad}} \mathcal{L}_{\text{grad}}^{\mathcal{V}_{\text{grad}}}(\hat{\mathbf{X}}^t, \mathbf{X}^t) \right], \quad (1)$$

where N_{rollout} denotes the number of rollout steps used during training, and $\mathbb{T}$ is the set of lead time steps from one to N_{rollout} . The subsets of prognostic variables $\mathcal{V}_* \subset \mathcal{V}$ are defined below together with each loss function $\mathcal{L}_*$. The hyperparameters λ_* , w_p , η_* , ζ , and ξ introduced below are specified as in Table S1. Note that, at training time, each prognostic variable is standardized; however—except for the first (Fourier-amplitude) term in Eq. (8)—the loss formulations below are invariant to this standardization.

2.3.1 MSE Loss $\mathcal{L}_{\text{mse}}$

For a subset of variables $\mathcal{V}_{\text{mse}} \equiv \{\text{SAT, SLP, TPW, DLR, DSR, } T_p, Q_p, Z_p \mid p = 1000, 850, 500, 300\}$, we apply the conventional MSE loss defined as

$$\mathcal{L}_{\text{mse}}^{\mathcal{V}_{\text{mse}}} = \sum_{\alpha \in \mathcal{V}_{\text{mse}}} s_\alpha w_\alpha \left\langle \left(\hat{X}_\alpha - X_\alpha \right)^2 \right\rangle_{\text{domain}} \\ = \sum_{\alpha \in \mathcal{V}_{\text{mse}}} s_\alpha w_\alpha \frac{1}{NM} \sum_{n,m} \left(\hat{X}_{\alpha nm} - X_{\alpha nm} \right)^2, \quad (2)$$

where $\langle \cdots \rangle_{\text{domain}}$ denotes spatial averaging and $X_\alpha \in \mathbb{R}^{N \times M}$ is the field of variable α extracted from $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times N \times M}$. The coefficient s_α is the inverse variance of temporal differences for variable α , and w_α is the variable-specific weight listed in Table 1. This formulation follows Lam et al. (2023) and Oskarsson et al. (2023).

2.3.2 Wind Loss $\mathcal{L}_{\text{wind}}$

For wind, instead of simply applying MSE loss to the u and v components, we decompose the error into directional and speed components, and then add penalties also on the associated divergence and curl fields. This design follows Sekiyama et al. (2023) and is intended to improve the physical fidelity of wind representations.

For each level $p \in \{\text{sfc, 1000, 850, 500, 300}\}$, let the horizontal wind vector field be $\mathbf{v}_p \equiv (u_p, v_p) \equiv (X_\alpha)_{\alpha \in \{U_p, V_p\}}$ with prediction $\hat{\mathbf{v}}_p \equiv (\hat{u}_p, \hat{v}_p) \equiv (\hat{X}_\alpha)_{\alpha \in \{U_p, V_p\}}$. Then the wind loss is defined as

$$\mathcal{L}_{\text{wind}} = \sum_p w_p \left[\langle \text{CosDis}(\hat{\mathbf{v}}_p, \mathbf{v}_p) \rangle_{\text{domain}} + \eta_{\text{mag}} \langle [\text{MagDif}(\hat{\mathbf{v}}_p, \mathbf{v}_p)]^2 \rangle_{\text{domain}} \right. \\ \left. + \eta_{\text{div}} \langle [\text{DivDif}(\hat{\mathbf{v}}_p, \mathbf{v}_p)]^2 \rangle_{\text{domain}} + \eta_{\text{curl}} \langle [\text{CurlDif}(\hat{\mathbf{v}}_p, \mathbf{v}_p)]^2 \rangle_{\text{domain}} \right], \quad (3)$$

where

$$\text{CosDis}(\hat{\mathbf{v}}_p, \mathbf{v}_p) = \frac{1}{2} \left(1 - \frac{\hat{\mathbf{v}}_p \cdot \mathbf{v}_p}{\|\hat{\mathbf{v}}_p\| \|\mathbf{v}_p\|} \right), \quad (4)$$

$$\text{MagDif}(\hat{\mathbf{v}}_p, \mathbf{v}_p) = \|\hat{\mathbf{v}}_p\| - \|\mathbf{v}_p\|, \quad (5)$$

$$\text{DivDif}(\hat{v}_p, v_p) = \left(\frac{\partial u_p}{\partial x} + \frac{\partial v_p}{\partial y} \right) - \left(\frac{\partial \hat{u}_p}{\partial x} + \frac{\partial \hat{v}_p}{\partial y} \right), \quad (6)$$

$$\text{CurlDif}(\hat{v}_p, v_p) = \left(\frac{\partial v_p}{\partial x} - \frac{\partial u_p}{\partial y} \right) - \left(\frac{\partial \hat{v}_p}{\partial x} - \frac{\partial \hat{u}_p}{\partial y} \right). \quad (7)$$

Here, $\langle \cdot, \cdot \rangle$ denotes the inner product and $\| \cdot \|$ the vector length. The partial derivatives are discretized and Eqs. (4)–(7) are computed in a grid-point-wise manner.

2.3.3 FalFcl Loss $\mathcal{L}_{\text{falFcl}}$

For hourly precipitation $\mathcal{V}_{\text{falFcl}} = \{\text{R1h}\}$, blurry predictions often arise when using MSE. To address this issue, alternative loss functions have been proposed that can explicitly separate amplitude error and phase/spatial structural error, thereby recovering high-frequency patterns and avoiding blurry predictions (Yan et al. 2024; Subich et al. 2025). Following this strategy, we define the precipitation loss function as

$$\begin{aligned} \mathcal{L}_{\text{falFcl}}^{\mathcal{V}_{\text{falFcl}}} = & \sum_{\alpha \in \mathcal{V}_{\text{falFcl}}} \frac{1}{NM} \sum_{k,l} \left(\sqrt{\text{PSD}_{k,l}(\hat{X}_\alpha)} - \sqrt{\text{PSD}_{k,l}(\tilde{X}_\alpha)} \right)^2 \\ & + \zeta \sum_{\alpha \in \mathcal{V}_{\text{falFcl}}} \left\{ 1 - \frac{\sum_{k,l} \Re(\hat{X}_{\alpha kl} \mathcal{X}_{\alpha kl}^*)}{\sqrt{\sum_{k,l} \text{PSD}_{k,l}(\hat{X}_\alpha) \times \sum_{k,l} \text{PSD}_{k,l}(X_\alpha)}} \right\}, \end{aligned} \quad (8)$$

where $\tilde{X}_\alpha$ is the standardized field of X_α (mean–std normalization) and $\Re(\cdot)$ denotes the real part. The power spectral density (PSD) and discrete Fourier transforms are defined as follows:

$$\text{PSD}_{k,l}(X_\alpha) = |\mathcal{X}_{\alpha kl}|^2, \quad \text{PSD}_{k,l}(\hat{X}_\alpha) = |\hat{\mathcal{X}}_{\alpha kl}|^2, \quad (9)$$

$$\mathcal{X}_{\alpha kl} = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X_{\alpha nm} \exp\left(-2\pi i \left(\frac{kn}{N} + \frac{lm}{M}\right)\right), \quad (10)$$

$$\hat{\mathcal{X}}_{\alpha kl} = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} \hat{X}_{\alpha nm} \exp\left(-2\pi i \left(\frac{kn}{N} + \frac{lm}{M}\right)\right). \quad (11)$$

Yan et al. (2024) refer to the first term of Eq. (8) representing the amplitude error as the Fourier Amplitude Loss (FAL), and the second term representing phase/spatial structural error as the Fourier Correlation Loss (FCL). This combination is expected to mitigate MSE-induced blurring and thus sharpens precipitation structures.

2.3.4 Grad Loss $\mathcal{L}_{\text{grad}}$

In pressure and geopotential fields $\mathcal{V}_{\text{grad}} = \{\text{SLP}, Z_p(p = 1000, 850, 500, 300)\}$, undesired

jagged features may appear when plotted with contours. To suppress such artifacts, we introduce smoothness with respect to the gradient through the following loss function:

$$\begin{aligned} \mathcal{L}_{\text{grad}}^{\mathcal{V}_{\text{grad}}} = & \sum_{\alpha \in \mathcal{V}_{\text{grad}}} \left[\langle \text{GradCosDis}(\hat{X}_\alpha, X_\alpha) \rangle_{\text{domain}} \right. \\ & \left. + s_\alpha \xi \langle [\text{GradMagDif}(\hat{X}_\alpha, X_\alpha)]^2 \rangle_{\text{domain}} \right], \end{aligned} \quad (12)$$

where

$$\text{GradCosDis}(\hat{X}_\alpha, X_\alpha) = \frac{1}{2} \left(1 - \frac{\nabla \hat{X}_\alpha \cdot \nabla X_\alpha}{\|\nabla \hat{X}_\alpha\| \|\nabla X_\alpha\|} \right), \quad (13)$$

$$\text{GradMagDif}(\hat{X}_\alpha, X_\alpha) = \|\nabla \hat{X}_\alpha\| - \|\nabla X_\alpha\|. \quad (14)$$

Here, the gradient ∇ is discretized and Eqs. (13) and (14) are computed in a grid-point-wise manner. Grad loss does not constrain the actual magnitudes of the variables and can therefore be used together with MSE loss.

2.4 Training and Prediction

The RRJ-ClimCORE subset for 2019–2022 was split as follows: from 2019–2021 (36 months), 29 months were used for training and 7 for validation, selected nonconsecutively across seasons; the entire year 2022 was held out as an independent test set.

Training used distributed data parallelism (DDP) on eight 96-GB H100 GPUs (batch size 4), AdamW (learning rate 0.002), and mixed precision. The model was first trained for 500 epochs with single-step forecasts and subsequently fine-tuned for 100 epochs with 4-step rollouts.

Because our model is regional, boundary conditions are required during rollouts. At each time step, the 10 grid points inward from the lateral boundary of the 470×470 domain are overwritten with ground-truth RRJ-ClimCORE values, and the updated fields are then provided as inputs for the next forecast step. This use of prescribed lateral boundary updates from the reanalysis is an idealized configuration, and the resulting errors are not directly comparable to those of an operational system; operationally, these boundary values would be supplied by global forecasts. For the lower boundary, SST is prescribed and held fixed at its initial value during each rollout, with the initial SST fields taken from RRJ-ClimCORE.

3 Forecast Evaluation

We present statistical evaluations based on skill scores averaged over space and time. As an ablation study, we compare the proposed model trained with the loss functions described in Sect. 2.3 against four alternative models in which individual components of the loss were removed, yielding five models in total:

- **mse** uses MSE loss for all variables.
 - **wind** applies Wind loss to wind variables (U_p , V_p) and MSE loss to all other variables.
 - **wind+falfcl** applies Wind loss to wind variables, FalFcl loss to R1h, and MSE loss to the other variables.
 - **wind+grad** applies Wind loss to wind variables, both MSE and Grad loss to SLP and Z_p , and MSE loss to the remaining variables.
 - **all** uses the complete loss function defined in Sect. 2.3.
- Further details of the individual models are provided in Table S1. Throughout this section, RRJ-ClimCORE is used as the ground truth.

3.1 RMSE Evaluations

We evaluate RMSE against the ground truth for each variable. Figure 1 presents the RMSE for U_p , Z_p , and SLP, while results for the other variables are shown in Figs. S2 and S3. For reference, we also include a persistence baseline—i.e., a forecast that the current state persists so that future weather

equals the initial state—plotted alongside our models in Figs. S4 and S5.

Overall, models with tailored losses (notably **all** and **wind+grad**) tend to achieve smaller RMSE for pressure/geopotential and several wind fields (Fig. 1a–e), i.e., non-MSE loss designs can outperform the pure-MSE baseline on RMSE. This does not contradict the intended effects of our loss design. For SLP and Z_p , we retain an MSE term and add the Grad loss, which aligns the direction and magnitude of spatial gradients and thereby lowers RMSE. For winds, the Wind loss targets direction, speed, divergence, and curl, providing a more informative learning signal than component-wise MSE and sometimes yielding lower a posteriori RMSE for U_p and V_p . At the same time, we note exceptions: for upper-tropospheric winds (300 hPa; Fig. 1f) and some other variables (Figs. S2, S3), **all** underperforms **mse**, warranting further investigation. Finally, for precipitation (Fig. S2e), **mse** yields smaller RMSE than the FalFcl-based models (**wind+falfcl**, **all**), consistent with its field-smoothing behavior that mitigates the double-penalty. To evaluate the effectiveness of our models for precipitation, it is more appropriate to assess the occurrence of precipitation events than strict grid-point accuracy as measured by RMSE; we examine this in the next subsection.

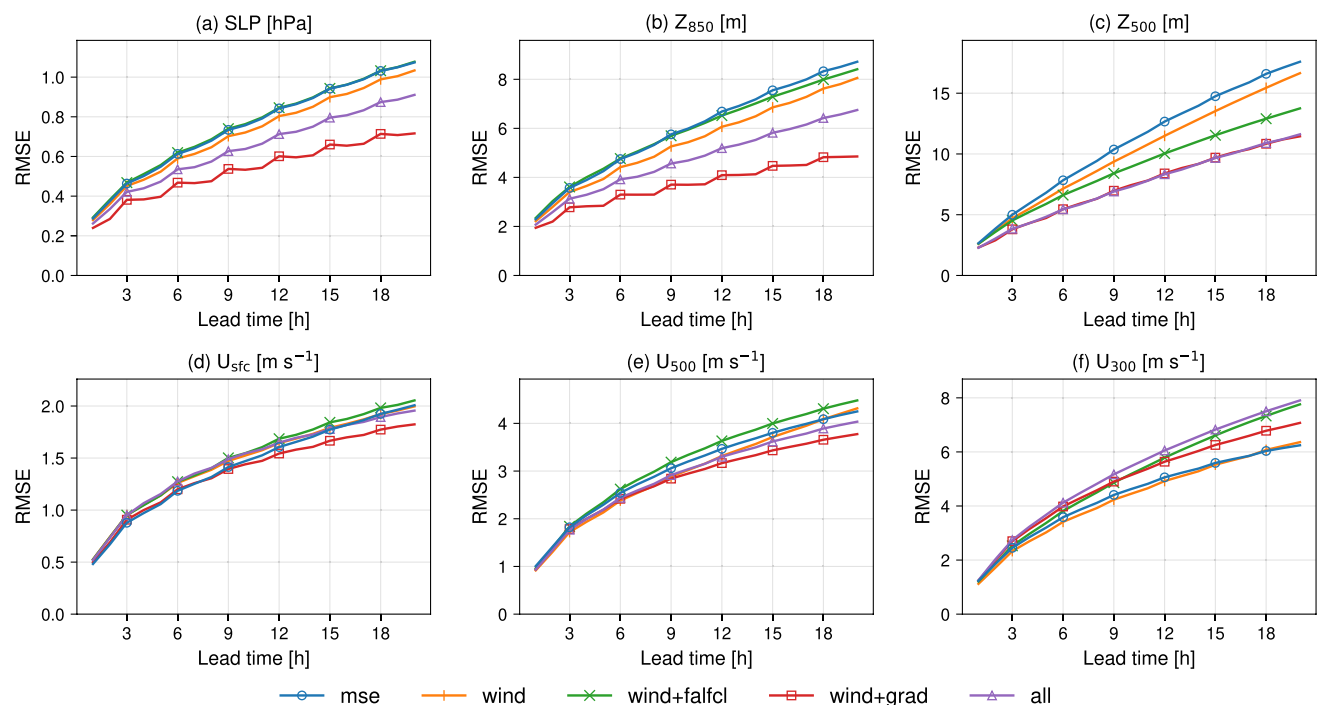


Fig. 1 RMSE of sea-level pressure, geopotential height, and wind forecasts from five models (blue: mse, orange: wind, green: wind+falfcl, red: wind+grad, purple: all) over a 12-month period in 2022, shown as

a function of forecast lead time. The variables are **a** SLP, **b** Z_{850} , **c** Z_{500} , **d** U_{sfc} , **e** U_{500} , and **f** U_{300}

3.2 Fuzzy Verification for Precipitation

For precipitation, we also conduct fuzzy verification (Ebert 2008), i.e., a coarser-scale evaluation that tolerates small space–time offsets. Specifically, for both the ground truth and forecasts, we define square domains with sides of 25 km and 100 km around each grid point, in each of which the neighborhood maximum of the 3-hour accumulated precipitation was evaluated. We then compare these box maxima with prescribed thresholds to form 2×2 contingency tables and compute categorical scores using pooled counts over all boxes and initial times. This approach assesses whether a heavy precipitation event occurs within a nearby window, accommodating typical displacement errors.

For completeness, we provide exact definitions of the categorical metrics used. For a given threshold and lead time, let TP (true positives/hits), FP (false positives/false alarms), FN (false negatives/misses), and TN (true negatives) denote the pooled counts from all boxes and initial times, and let $N_{\text{all}} = \text{TP} + \text{FP} + \text{FN} + \text{TN}$. The Equitable Threat Score (ETS) and Frequency Bias Index (FBI) are

$$\text{ETS} = \frac{\text{TP} - H_{\text{rand}}}{\text{TP} + \text{FP} + \text{FN} - H_{\text{rand}}},$$

$$H_{\text{rand}} = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN})}{N_{\text{all}}},$$

$$\text{FBI} = \frac{\text{TP} + \text{FP}}{\text{TP} + \text{FN}},$$

respectively. Here, larger ETS indicates higher skill after accounting for hits expected by random chance ($\text{ETS} = 0$ denotes no skill), and $\text{FBI} = 1$ indicates unbiased event frequency, with values < 1 and > 1 indicating under- and overforecasting, respectively.

Figure 2 summarizes ETS across lead times and box sizes for the five models. ETS improves most for **all** across the lead times and box sizes. By comparing across the models, we can evaluate relative contributions of individual loss components. Both **mse** and **wind** yield nearly identical ETS, while **wind+falcl** leads to a marked improvement, demonstrating that FalFcl loss plays a crucial role in precipitation prediction. Interestingly, **wind+grad** yields only modest improvements in ETS relative to **wind**, but when combined with FalFcl (i.e., **all**), the ETS increases substantially, suggesting that Grad and FalFcl losses have complementary effects.

Figure 3 provides the corresponding FBI evaluated over the same lead times and box sizes. The **mse**, **wind**, and **wind+grad** models overall yield underforecasting, especially for heavy rainfall. The **wind+falcl** model partially mitigates this bias, but underforecasting remains. By contrast, **all** maintains FBI values close to 1 even for intense precipitation during the 0–3 h forecast range (Fig. 3a, b). During the 6–9 h forecast range (Fig. 3c, d), **all** also yields substantial improvement despite a slight underforecasting bias. These results imply that **all** is able to predict rainfall amounts comparable to the ground truth. Histograms of precipitation frequency at the grid-point scale in Fig. S6 further support these findings. Comparing Fig. 3a, c with Fig. 3b, d,

Fig. 2 ETS of precipitation for five models (blue: mse, orange: wind, green: wind+falcl, red: wind+grad, purple: all) evaluated over a 12-month period in 2022. The ETS is computed with respect to the maximum 3-hour accumulated precipitation within each box domain. Results are shown for different forecast lead times and box sizes: **a** 0–3 h, 25 km; **b** 0–3 h, 100 km; **c** 6–9 h, 25 km; and **d** 6–9 h, 100 km

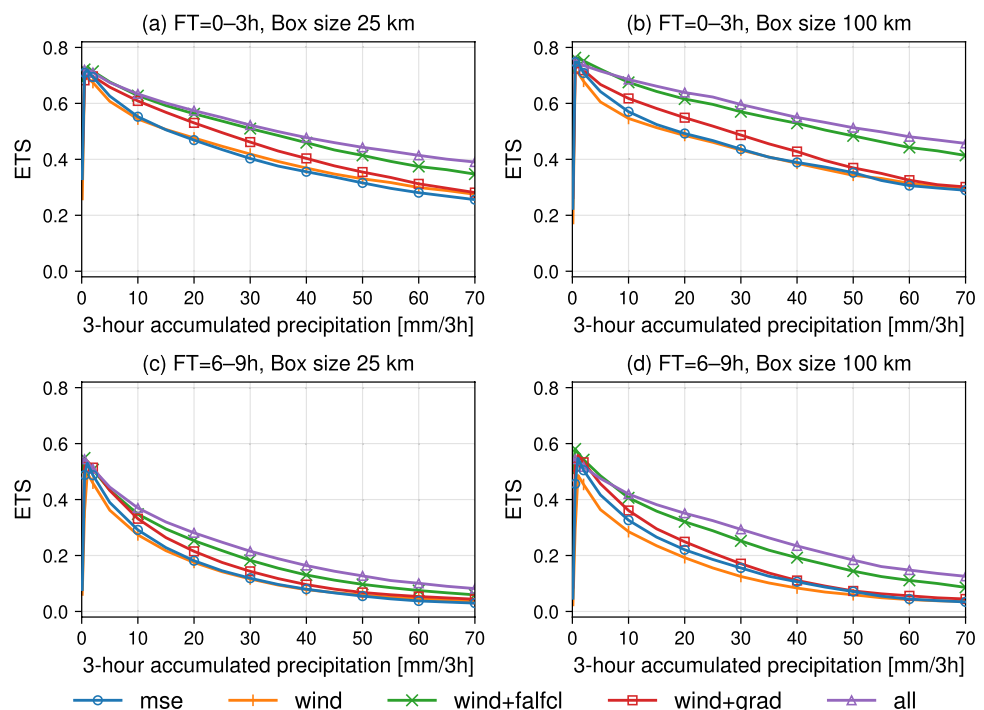
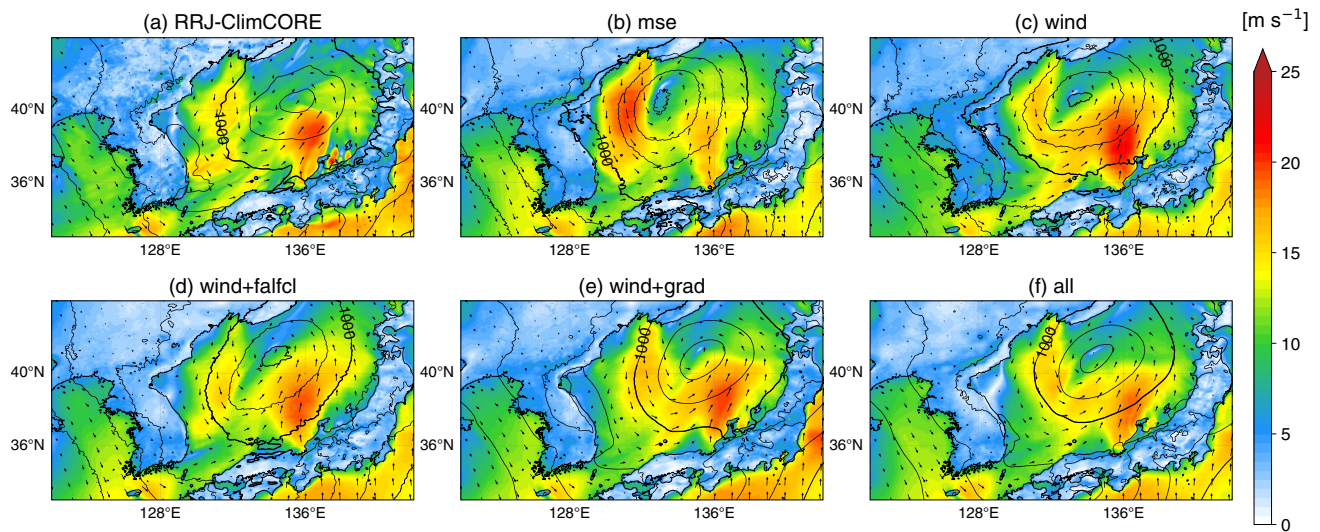
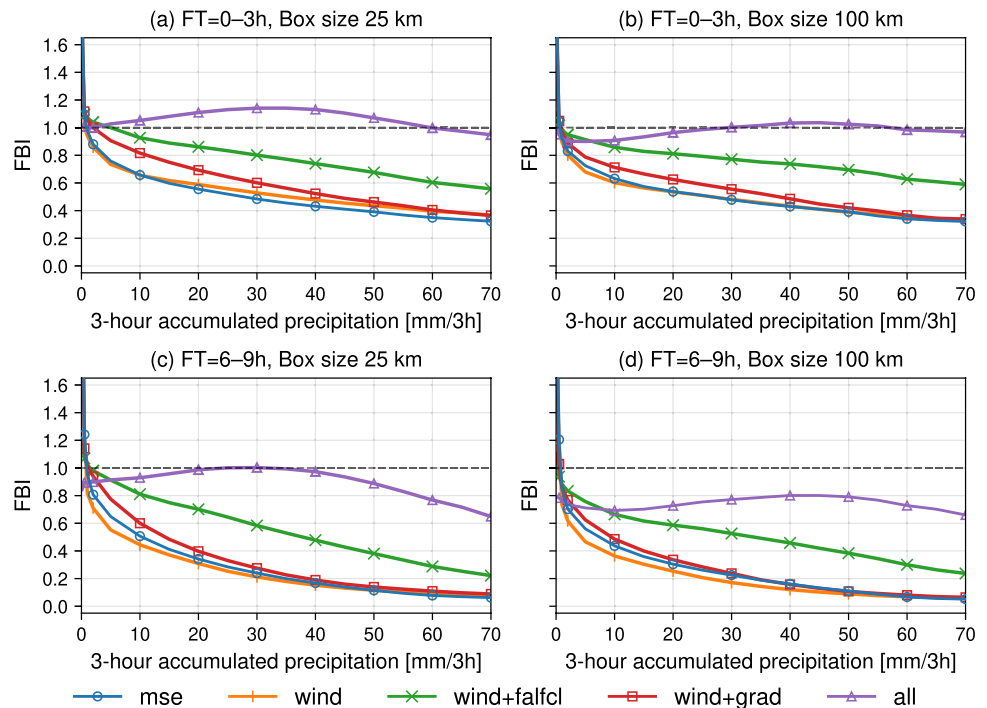


Fig. 3 Same as Fig. 2, but for FBI**Fig. 4** 10-m wind speed (color), 10-m wind vectors (arrows), and SLP (contours) at 0900 UTC on 26 March 2022, together with 12-h forecasts initialized at 2100 UTC on 25 March 2022 and valid at 0900

UTC on 26 March 2022. Contour intervals are 4 hPa, with bold lines every 20 hPa. Panels show **a** RRJ-ClimCORE (ground truth), **b** mse, **c** wind, **d** wind+falfcl, **e** wind+grad, and **f** all

the FBI tends to be larger for the smaller boxes (25 km) than for the larger ones (100 km). A plausible interpretation is that **all** may produce heavy-rainfall features that are somewhat more spatially concentrated than in the ground truth, which would raise counts within smaller boxes relative to larger ones.

4 Case Studies

4.1 Extratropical Cyclone

On 26 March 2022, an extratropical cyclone moved eastward over the Sea of Japan, with strong southerly winds especially where the flow was channeled through narrow low-elevation gaps along the coast, as captured by the regional reanalysis. Figure 4 shows 12-h forecasts (valid at 0900 UTC on 26 March 2022) of 10-m wind speed,

wind vector, and SLP from the individual models, together with the ground truth. At that time, the cyclone was in its developing stage, with intense southerly winds evident south of the cyclone center and strong northerly winds to the west (Fig. 4a). The **wind** model (Fig. 4c) forecasts the strength and distribution of these strong winds better than **mse** (Fig. 4b). The **wind+grad** model (Fig. 4e) not only forecasts the wind field realistically but also yields a SLP distribution around the cyclone center that is closer to the ground truth. In addition, thanks to the Grad loss, it yields smoother isobars than undesirable jagged lines seen in **wind** and **wind+falfcl** (Fig. 4c, d). The **all** model (Fig. 4f) inherits these favorable characteristics of **wind+grad**.

4.2 Typhoon Nanmadol

Typhoon Nanmadol, which formed south of Japan on 14 September 2022, made landfall in southwestern Kyushu, bringing record-breaking rainfall and destructive winds. Figure 5 compares 6-hour forecasts (valid at 0900 UTC on 18 September) of R1h and SLP from the individual models with the ground truth. In the ground truth (Fig. 5a), heavy precipitation was analyzed over Kyushu, which—as discussed later—was of orographic origin; however, **mse**

(Fig. 5b) seriously underestimates this heavy rainfall. By contrast, **wind+falfcl** and **all** (Fig. 5d, f), which incorporate the FalfCl loss, mitigate this underestimation and, despite minor spatial displacements, forecast heavy precipitation comparable to the ground truth. We also note that **wind** (Fig. 5c) produces heavy rainfall in this event. Based on the fuzzy-verification results, this does not appear to be a universal outcome for **wind**; thus, there may be case-specific factors at play. In what follows, we focus on the orographic nature of this rainfall event and discuss the implications.

The top row of Fig. 6 shows the ground truth R1h (an enlarged view of Fig. 5a) together with a topographic map of the region and wind vectors at 850 hPa. The dashed box highlights mountainous areas. The intense precipitation occurred on the windward slope of the local mountain range, where the southeasterly winds blew, indicating that the rainfall was induced by orographic lifting of a moist airflow. Consistently, RRJ-ClimCORE shows 850-hPa convergence on the windward side of the mountain range and divergence on its leeward side (Fig. 6a). Figure 6b–f present the corresponding 6-h forecast fields for the models. In **mse** (Fig. 6b), the convergence–divergence structure is indistinct. In contrast, **wind** (Fig. 6c) produces a clearer convergence–divergence pattern, more closely resembling

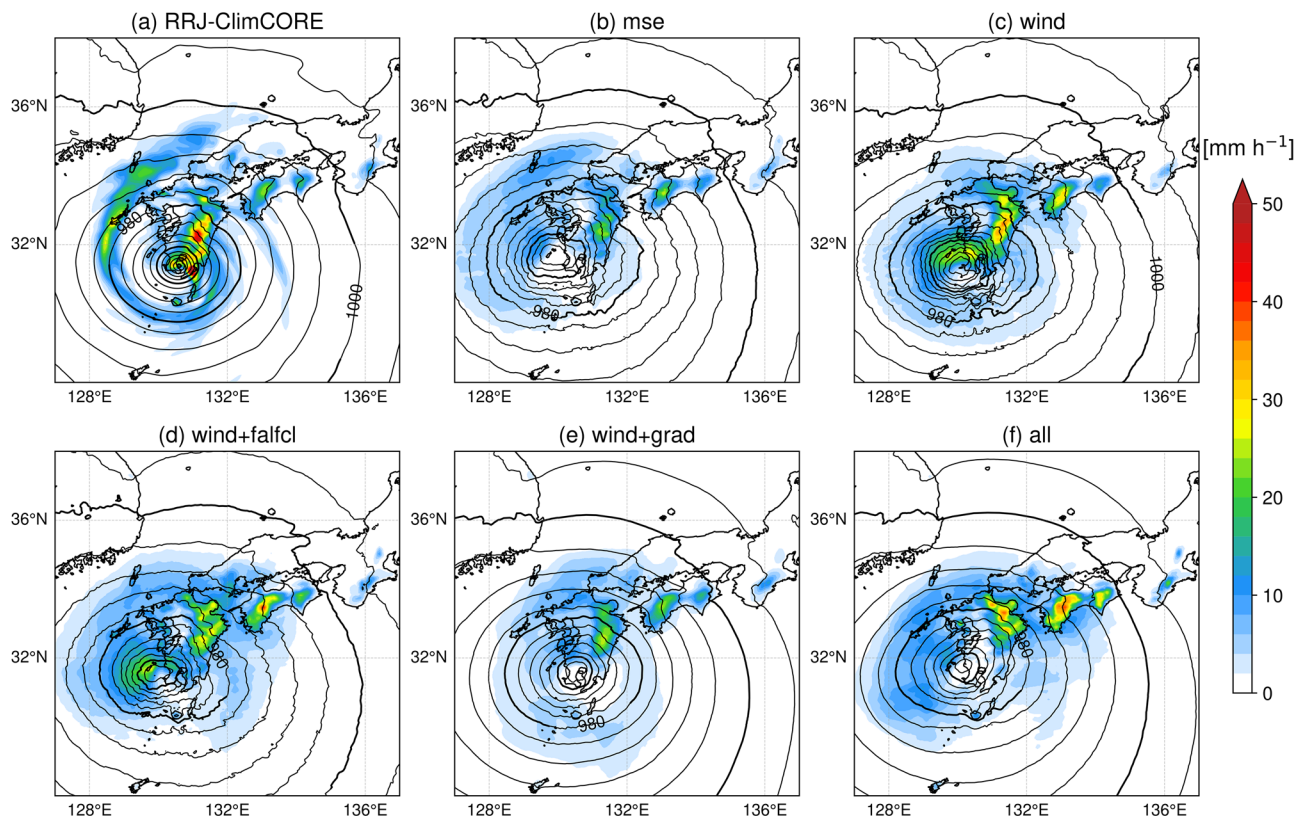


Fig. 5 R1h (color) and SLP (contours) at 0900 UTC on 18 September 2022, together with 6-h forecasts initialized at 0300 UTC on the same day and valid at 0900 UTC. Contour intervals are 4 hPa, with bold

lines every 20 hPa. Panels show **a** RRJ-ClimCORE (ground truth), **b** mse, **c** wind, **d** wind+falfcl, **e** wind+grad, and **f** all

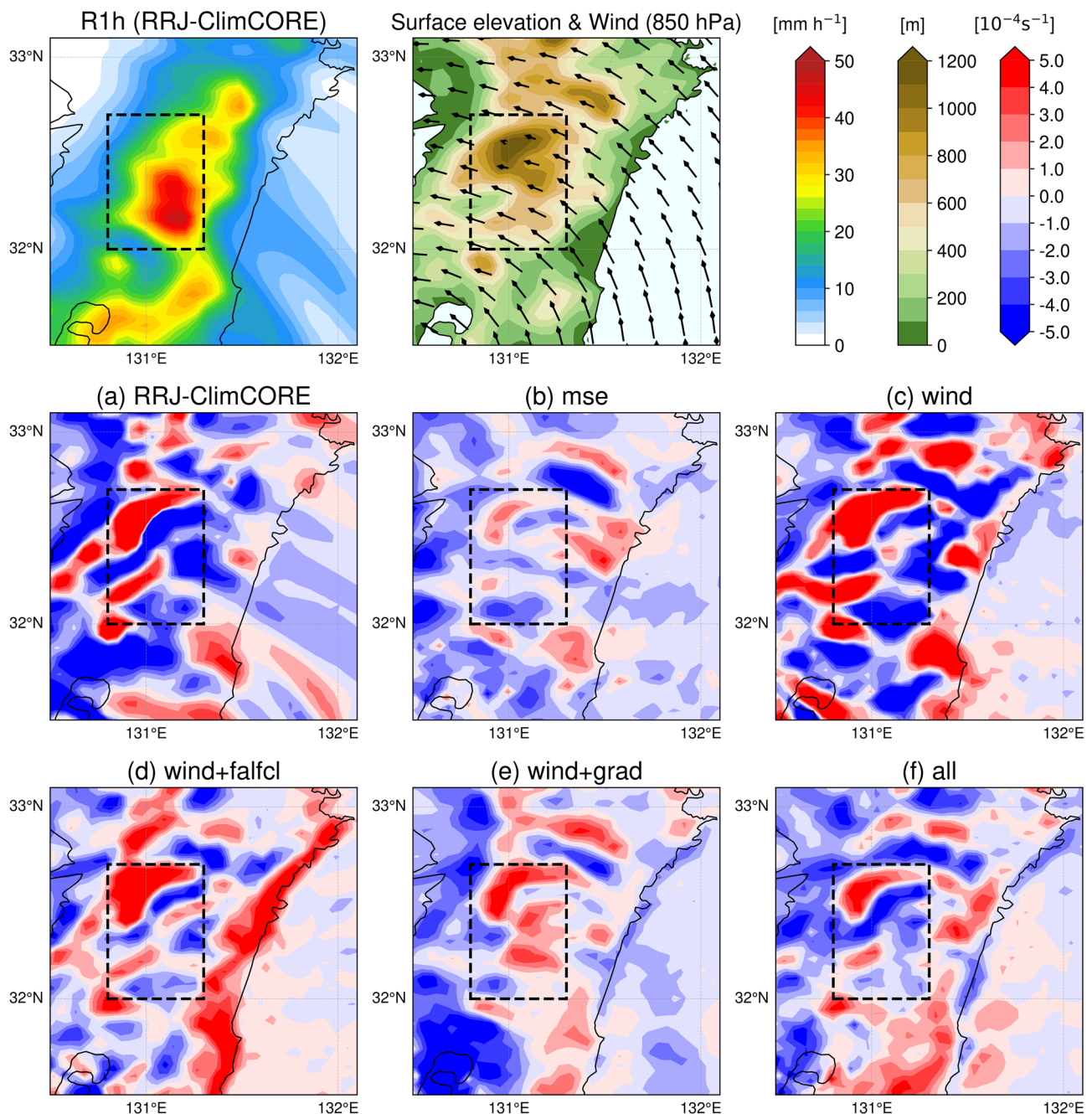


Fig. 6 Top row: R1h from RRJ-ClimCORE at 0900 UTC on 18 September 2022, shown as an enlarged view of Fig. 5a, along with RRJ-ClimCORE 850-hPa wind vectors (arrows) at the same time together with surface elevation (color). Middle and bottom rows: horizontal divergence at 850 hPa (color) from 6-h forecasts initialized at 0300

UTC and valid at 0900 UTC on the same day, for **a** RRJ-ClimCORE (ground truth), **b** mse, **c** wind, **d** wind+falfcl, **e** wind+grad, and **f** all. In all panels, regions of heavy precipitation associated with orography are highlighted by black dashed boxes

the ground truth. This indicates that improved wind representation around the mountain can enhance the forecast precipitation intensity. While MLWP does not explicitly represent physical causality, these results imply that the model learned clearer correlations among wind, precipitation, and topography.

Finally, we also examined the forecast skill for typhoon intensity. Table S2 summarizes comparisons of central SLP and maximum surface wind speed among the best track, reanalyses, our models, and AIFS (Lang et al. 2024), a global MLWP model trained on ERA5. Both ERA5 and AIFS show substantially smaller values than the best track and RRJ-ClimCORE for both variables. In contrast, even

mse improves upon these baselines, likely owing to the high fidelity of RRJ-ClimCORE in reproducing extreme events. Moreover, further improvements are evident with **wind**, which provides the most accurate intensity forecasts in this case. Meanwhile, **all** does not outperform **mse**, indicating that the effectiveness of individual loss functions may depend on the specific event and warrants further investigation.

5 Conclusion

We developed a regional MLWP model for Japan using regional reanalysis (RRJ-ClimCORE) and tailored variable-specific loss functions. Fuzzy verification, RMSE, and case studies show complementary gains: the FalFcl loss sharpens precipitation, while the Wind loss improves winds and associated orographic rainfall, and the Grad loss reduces jagged pressure/height contours, jointly improving extreme-event forecasts.

Despite these advances, our study has limitations. The model relies on externally supplied lateral boundaries (set to the ground truth in this study), so sensitivity to boundary quality—especially at longer lead times—remains to be quantified, and fair NWP comparisons will require matched boundary conditions. While the Grad loss mitigated jagged contours in this architecture, other backbones may produce different artifacts (e.g., grid-like artifacts reported for generic Vision Transformer architectures Yang et al. 2024), implying that the optimal loss design could change. Our results also indicate room for improvement in upper-level winds and tropical cyclone intensity, highlighting the need to elucidate the characteristics of each loss function and pursue corrective measures.

Future work will focus on conducting a detailed quantitative examination of the contributions of the individual loss function components and clarifying the meteorological mechanisms underlying the forecast improvements. Another important direction is to extend the training to longer and more diverse regional datasets and to conduct broader case studies, including systematic comparisons with operational forecasting models.

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Author Contributions H.I. implemented the model, conducted evaluation experiments, and wrote the majority of the manuscript; T.T.S. contributed essential methodological ideas, conducted the Table S2

experiment, and provided technical advice; T.M. prepared and curated the dataset, formatted it for efficient use in the experiments, provided technical comments, and supervised H.I.; K.K. contributed to interpretation and discussion, offered strategic guidance from an operational-meteorology perspective, and facilitated the collaboration by engaging T.T.S.; H.N. contributed to interpretation and discussion, provided overall guidance, managed the project, and supervised H.I.; all authors discussed the results, contributed to revision, and approved the final version.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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